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What a Transformer Learns When It Adapts to Unknown Physics In-Context: Probing, Patching and Controlling

nonberw HTTPS://GITHUB.COM/NONBERW/SUSS
Independent researcher

Abstract

Sequence models that adapt to unknown dynamics from their own history are common in robot learning, but what they compute is unclear. We study small transformers trained from scratch, in a pre-registered design with exact optimal references and causal interventions, in the smallest settings where the question has exact answers: linear systems with unknown parameters, one-degree-of-freedom MuJoCo systems with contact, and linear adaptive control. The predictors are Bayes-optimal while data are scarce, then settle at an approximation floor (1.59× the Bayes excess risk). Our pre-registered mechanism, a linearly readable and causally used parameter estimate in the residual stream, is rejected (probe R^2=0.19; steering and patching have no effect). Exploratory analyses locate the sufficient statistics of the context in the attention key–value memory of the last two blocks: they are read from it with R^2=0.92, against 0.68–0.71 for the best residual-stream readout, and replacing only those keys and values with another episode's transfers 0.91 – -0.94 of its behaviour. Physical parameters in MuJoCo are also more readable from the key–value memory than from the residual stream, by margins that vary with the system. A controller trained by imitation and then on closed-loop cost excites the plant without an identification objective, though less than dithered certainty equivalence, and is competitive in median cost but fails on 10.0% of plants, almost all open-loop unstable, against 1.4%. Code, pre-registration and every number are public.

Keywords: in-context learning, adaptive control, system identification, dual control, mechanistic interpretability

1. Introduction

A robot picking up an unfamiliar object, or a legged robot stepping onto ice, must adapt to physics it does not know from a few seconds of interaction. A popular recipe trains a sequence model across randomised physics and lets it adapt *in context*, from its own history, with no gradient steps at deployment (Kumar et al., 2021; Duan et al., 2016; Forgione et al., 2023; Busetto et al., 2024). These models work, but it is not clear what they compute. Do they identify the plant and then plan? Do they implement a classical estimator? Do they explore?

For linear-Gaussian dynamics these questions have exact answers, so they can be tested. Transformers trained on random linear systems are known to *behave* like the Kalman filter or least squares (Balim et al., 2023; Akram and Vikalo, 2026). They are also known to be *able to represent* such estimators (Goel and Bartlett, 2024; Cole et al., 2025; Vlădărean et al., 2026). What has not been measured is what is inside a trained in-context system identifier, or what an in-context *controller* does about the dual effect. The second question matters because, under exploration-free feedback, closed-loop data do not identify the plant (Feldbaum, 1960; Åström and Wittenmark, 1995; Ljung, 1999). Akram and Vikalo (2026) list the first question as future work.

We study both questions in a pre-registered design. Hypotheses, metrics, seeds and tests were committed to the repository before the main experiments, and every deviation is logged. We deliberately stay in the smallest settings in which the questions have exact answers: linear systems

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with unknown parameters, one-degree-of-freedom MuJoCo systems with contact (Todorov et al., 2012), and linear adaptive control. The aim is a clean mechanism, not a deployable controller. Our contributions:

- **What the trained predictor computes (§5).** It is Bayes-optimal while data are scarce, and then it settles at an approximation floor. Our pre-registered mechanism, a linearly readable and causally used parameter estimate in the residual stream, is *rejected*: the probe reaches R^2=0.19, and steering and subspace patching have no effect. Exploratory analyses instead localise the context information in the *key–value memory* of the last two blocks: replacing only those keys and values with another episode's transfers $0.91--0.94$ of its behaviour (first block alone: 0.07–0.16), and the predictor's query Jacobian, its functional estimate, matches the Bayes estimate (R^2=0.92).
- **Contact physics (§6).** On a pushed block with Coulomb friction the transformer beats linear-in-parameters physics-feature RLS by $3.2\times$, and on a pendulum it is on par with RLS, which there has the true model class. Physical parameters are weakly readable from activations ($1/m$ on the pendulum: R^2=$0.32--0.44$) and more readable from the key–value memory ($0.97--0.98$); on the block the gap is smaller (μ: 0.68–0.72 vs. 0.51–0.52).
- **In-context control (§7).** A transformer controller trained by imitation and then on cost excites the plant without being asked to (4.20 nats of information more than exploration-free certainty equivalence, about half of it learned from the cost), but less than dithered certainty equivalence (-1.41 nats). Its median cost is comparable to dither and Thompson sampling, while its failure rate is 10.0% against 1.4%, concentrated on open-loop-unstable plants, and cost fine-tuning makes those failures more severe. It keeps its estimate of the actuation matrix in the same key–value memory.
- **Theoretical background (§3).** We restate, for in-context predictors under closed-loop data, the Bayes risk of prediction with unknown parameters, its $np\sigma^2/t$ law, and closed-loop non-identifiability in the form "information along unexcited directions equals excitation energy". These are standard results; they turn the questions above into measurable gaps.

2. Related work

In-context learning of function classes. Transformers trained on random regression tasks match least squares (Garg et al., 2022), interpolate between ridge and OLS with intermediate quantities decodable by probes in static regression (Akyürek et al., 2023), can implement GD (von Oswald et al., 2023; Ahn et al., 2023; Zhang et al., 2024), higher-order iterations (Fu et al., 2023) and algorithm selection (Bai et al., 2023), and approximate posterior predictives (Müller et al., 2022; Raventós et al., 2023). We move these questions to closed-loop dynamical data, where the regressors depend on the past and on the policy, and where the dual effect has no i.i.d. analogue.

In-context learning of dynamical systems. Meta-trained transformers predict, filter and control unseen systems (Forgione et al., 2023; Piga et al., 2023; Busetto et al., 2023; Balim et al., 2023; Akram and Vikalo, 2026; Busetto et al., 2024). Constructions and approximation theory exist for Kalman filtering (Goel and Bartlett, 2024), least-squares learning of LDS (Cole et al., 2025) and single attention layers (Vlădărean et al., 2026). All of these are behavioural or representational; none intervenes on a trained model.

Adaptation from history, in-context RL, fast weights. RMA (Kumar et al., 2021) regresses a history encoding onto privileged physical parameters; probing such latents gives mixed results (Chamachot, 2026; Zhang et al., 2026). Meta-RL agents can learn to explore (Duan et al., 2016; Zintgraf et al., 2020), algorithm distillation imitates learning histories (Laskin et al., 2023), and in-context RL algorithms are representable in tabular MDPs (Liang and Lai, 2026). In adaptive LQR, certainty equivalence with dithering is rate-optimal (Mania et al., 2019; Simchowitz and Foster, 2020); these methods are our baselines. We use linear probes (Alain and Bengio, 2016) and activation patching (Zhang and Nanda, 2024; Heimersheim and Nanda, 2024), always paired with interventions, and the view of attention as reading and writing fast weights (Ba et al., 2016; Schlag et al., 2021; Dai et al., 2023) motivates our key–value analysis.

3. Theoretical background

The results of this section are standard (Bayesian ridge regression, closed-loop identifiability, and attention as fast weights), restated for in-context predictors under closed-loop data. We include short proofs for completeness and claim no novelty; their role is to turn the questions of §1 into measurable gaps.

Setting. A plant $x_{s+1} = \theta z_s + w_s$ with regressor $z_s = (x_s, u_s) \in \mathbb{R}^p$, unknown $\theta = [A\ B] \in \mathbb{R}^{n\times p}$, and noise $w_s \sim \mathcal{N}(0, \sigma^2 I_n)$ i.i.d. and independent of everything else. The rows θ_i of θ have independent priors $\mathcal{N}(0, \Lambda_0^{-1})$. Inputs are chosen by an arbitrary *non-anticipating* policy: u_s is a measurable function of the past data $\mathcal{D}_s = (x_0, u_0, \dots, x_s)$ and of independent randomness, so it does not depend on θ except through $\mathcal{D}_s$. An *in-context predictor* is any map $\hat{y}_t = f_t(\mathcal{D}_t, u_t)$, of which a causal transformer is one example. Write $S_t = \sum_{s<t} z_s z_s^\top$ and $C_t = \sum_{s<t} z_s x_{s+1}^\top$.

Theorem 1 (Bayes risk of in-context prediction; Bayesian ridge regression) *Under the untruncated Gaussian prior, for every in-context predictor and every t,*

$$\mathbb{E}\,\|\hat{y}_t - \theta z_t\|^2 \;\geq\; \mathcal{R}_t^\star := n\sigma^2\,\mathbb{E}\big[z_t^\top(\sigma^2\Lambda_0 + S_t)^{-1} z_t\big], \tag{1}$$

with equality iff $\hat{y}_t = C_t^\top(\sigma^2\Lambda_0 + S_t)^{-1} z_t$ almost surely. This is ridge regression, i.e. RLS initialised at the prior.

Proof Appendix B (Gaussian posterior with precision $\Lambda_0 + \sigma^{-2} S_t$, then bias–variance). ■

Corollary 2 (A parameter-free context-length law) *Fix θ and suppose that, conditionally on θ: (a) $S_t/t \to \Sigma_\theta \succ 0$ almost surely; (b) $\mathbb{E}[z_t z_t^\top \mid \theta] \to \Sigma_\theta$; (c) the families $\{t\, z_t^\top(\sigma^2\Lambda_0 + S_t)^{-1} z_t\}_t$ and $\{\|z_t\|^2\}_t$ are uniformly integrable. Then the conditional Bayes risk satisfies $t\,\mathcal{R}_t^\star(\theta) \to n\,p\,\sigma^2$. If, in addition, $\sup_t t\,\mathcal{R}_t^\star(\theta)$ is integrable over the prior, then $t\,\mathcal{R}_t^\star \to np\sigma^2$. The limit does not depend on the prior, on θ, or on the input distribution, because $\mathrm{tr}(\Sigma_\theta^{-1}\Sigma_\theta) = p$ whatever Σ_θ is. Conditions (a)–(b) are standard for stable plants driven by i.i.d. inputs (as in Experiment A); we check the limit numerically rather than verify (c).*

Proof Appendix B. ■

Theorem 1 turns "is the transformer Bayes-optimal?" into a measurable gap. Corollary 2 gives a target curve $\mathcal{R}_t^\star \approx np\sigma^2/t$ that involves no fitted constants. (In Experiment A, the prior is truncated

to stable A. The exact optimum is then the truncated posterior mean, which we compute by Monte Carlo. The truncation only matters at short contexts.)

Proposition 3 (Closed-loop non-identifiability; the dual effect) *Let the policy be $u_s = Kx_s + \varepsilon_s$ for a fixed $K \in \mathbb{R}^{m\times n}$ and an excitation sequence ε_s. For any $b \in \mathbb{R}^m$, let $v_b = (-K^\top b,\ b) \in \mathbb{R}^p$. Then for all t,*

$$v_b^\top \Sigma_t^{-1} v_b \;=\; v_b^\top \Lambda_0 v_b + \sigma^{-2} \textstyle\sum_{s<t} (b^\top \varepsilon_s)^2. \tag{2}$$

In particular, without excitation ($\varepsilon \equiv 0$), the posterior precision of every row along v_b stays at its prior value for every context length. By Cauchy–Schwarz, the posterior variance satisfies $v_b^\top \Sigma_t v_b \geq \|v_b\|^4/(v_b^\top \Lambda_0 v_b)$ for every t. This bound does not depend on t, and it equals the prior variance when Λ_0 is isotropic. So the combination $\theta v_b = (B - AK^\top)b$ is never learned. On-policy queries are unaffected, since $v_b^\top z_t = 0$ along the trajectory; what is lost is exactly the knowledge needed to act differently*, which is the dual effect.*

Proof $v_b^\top z_s = -b^\top K x_s + b^\top(Kx_s + \varepsilon_s) = b^\top \varepsilon_s$, so $v_b^\top S_t v_b = \sum_{s<t}(b^\top \varepsilon_s)^2$. Substitute into $\Sigma_t^{-1} = \Lambda_0 + \sigma^{-2} S_t$. The variance bound follows from $(v^\top v)^2 \leq (v^\top \Sigma v)(v^\top \Sigma^{-1} v)$. ■

The extreme case is $K = 0$ with a prior mean of zero for B: a certainty-equivalent controller then outputs $u \equiv 0$ forever and never learns anything about B, which Experiment C shows exactly. A good adaptive controller must *spend* control effort on excitation. We measure the information gathered as $\log\det(\Lambda_0 + \sigma^{-2} S_8)$ with $\Lambda_0 = I$; note that even $u \equiv 0$ yields a non-zero value, from the state block of S_8, so the relevant comparison is between controllers, not against zero.

Remark 4 (Where an estimate can live without appearing in any activation) *Consider a softmax head with keys k_s, values v_s and query q, and suppose its scores are small, $|q^\top k_s|/\sqrt{d_h} \ll 1$. This is the near-uniform regime, with normalised attention entropy close to 1. Expanding the softmax to first order,*

$$\textstyle\sum_s \mathrm{softmax}_s\, v_s \;\approx\; \bar{v} + \frac{1}{\sqrt{d_h}}\, \widehat{\mathrm{Cov}}(v,k)\, q, \qquad \widehat{\mathrm{Cov}}(v,k) = \frac{1}{t+1}\textstyle\sum_{s\leq t}(v_s - \bar{v})k_s^\top. \tag{3}$$

The head is an affine map of the query whose matrix is the key–value covariance ("fast weights" (von Oswald et al., 2023)), spread over all context positions. So an estimate can be applied to the query, showing up in the Jacobian $\partial \hat{y}_t/\partial z_t$, without ever being a vector in the residual stream: a residual probe for $\hat{\theta}$ can fail while a probe on $\widehat{\mathrm{Cov}}(v,k)$ and patching of keys and values succeed. Section 5 tests both.

Appendix A gives a two-layer linear-attention construction that places the exact statistics (S_t, C_t) in the residual stream; it motivated the pre-registered "statistics first, solve later" prediction (H2c).

4. Experimental design

Pre-registration and statistics. Hypotheses, metrics, seeds, thresholds and tests were committed before the main experiments (`PREREGISTRATION.md`, commit `d134c5a`); deviations are in Appendix C. Confirmatory claims use a hierarchical bootstrap (seeds, then episodes; 10^4 resamples) with Holm correction over the 13 confirmatory tests. Every model is a GPT-style causal decoder (pre-LN, d=128, 4 layers, 4 heads, 0.8M parameters) trained from scratch with three seeds on a 4 GB laptop GPU.

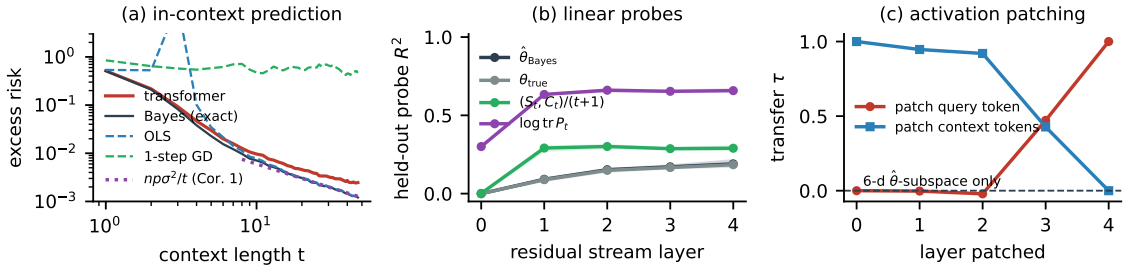


Figure 1: **A, pre-registered analyses** (3 seeds). (a) Excess risk vs. context length. The transformer tracks the exact Bayes predictor while data are scarce, then flattens at an approximation floor. The Bayes curve follows the parameter-free law of Cor. 2. (b) Held-out linear probes of the residual stream (band: range over seeds). The estimate $\hat{\theta}_t$ is barely decodable; the posterior uncertainty is. (c) Activation patching of episode A into B. Patching the query token transfers A's behaviour only from layer 3 on. Patching the 6-d $\hat{\theta}$-probe subspace transfers nothing (dashed).

A: in-context prediction of linear systems. $x \in \mathbb{R}^2$, $u \in \mathbb{R}$ (p=3), σ=0.1, prior $A_{ij} \sim \mathcal{N}(0, 0.4^2)$, $B_i \sim \mathcal{N}(0, 0.5^2)$ truncated to $\rho(A) < 0.95$, inputs $u_t \sim \mathcal{N}(0, 1)$, T=48; token t is (x_t, u_t), target x_{t+1}; fresh batches for 20k steps. References: the *exact* Bayes predictor (truncated posterior mean by Monte Carlo), ridge/RLS, OLS, tuned one-step GD (von Oswald et al., 2023), and zero. Excess risk is measured against the noiseless mean θz_t.

B: MuJoCo physics. *Pendulum*: point mass on a 0.5 m rod, $m \sim U[0.5, 2]$, damping $c \sim U[0.02, 0.3]$, $g \sim U[4, 16]$, random torques; regression on $(u, \dot{q}, \sin q, 1)$ contains the true model class. *Pushed block*: box on a plane, Coulomb friction $\mu \sim U[0.1, 1]$, $m \sim U[0.5, 3]$, random forces; about half the steps are stuck, so stick–slip lies outside any linear-in-parameters model. *Bouncing ball* (exploratory, added later): m, g and the contact damping ratio unknown. Control step 0.05 s, unobserved process noise, 60k training episodes, T=48. Baselines: simulation with the true or nominal parameters, physics-feature RLS, a no-context MLP, and on the block a structured identifier with the exact simulator.

C: in-context adaptive control. $A = 0.9I + \mathcal{N}(0, 0.25^2)$ truncated to $\rho(A) < 1.1$ (31% of plants are open-loop unstable), $B = r(\cos\phi, \sin\phi)$ with ϕ uniform (actuation direction and sign unknown), $J = \sum_{t<48} \|x_t\|^2 + 0.1u_t^2 + \|x_{48}\|^2$, $|u| \le 5$. The transformer reads (x_t, u_{t-1}) and outputs u_t; it never sees θ and has no identification loss. Baselines, tuned on separate plants: oracle finite-horizon LQR, the best constant gain, certainty equivalence (CE), CE with decaying dither (Simchowitz and Foster, 2020), explore-then-commit, and Thompson sampling. Test plants and noise are shared across controllers.

5. Experiment A: what the in-context predictor computes

Behaviour: Bayes-optimal while data are scarce, then an approximation floor. The trained predictor is much closer to the exact Bayes predictor than to OLS or to one GD step. The mean algorithmic distance to OLS exceeds the distance to Bayes by 0.271 [0.181, 0.379], and the distance

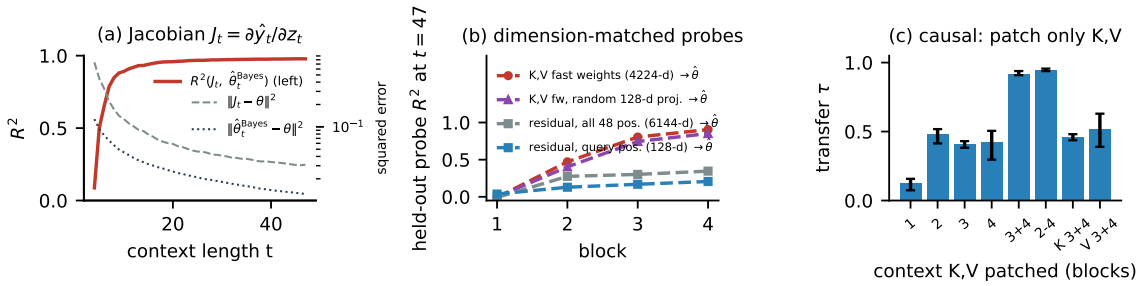


Figure 2: **A, exploratory follow-up: where the estimate lives** (3 seeds). (a) The predictor's Jacobian with respect to its query, $J_t{=}\partial\hat{y}_t/\partial z_t$, matches the Bayes estimate. Its error to the truth (dashed) tracks the Bayes estimate's error (dotted). (b) Probes for $\hat{\theta}_{47}$ by block: projected to 128 dimensions, the key–value fast weights still beat the residual stream over all 48 positions (Table 4 adds the sufficient statistics). (c) Patching *only* the context keys and values of blocks 3–4, leaving every activation of the query untouched, transfers over 90% of the other episode's behaviour. Keys or values alone transfer about half.

to GD exceeds it by 0.552 [0.496, 0.616] (H1a, H1b; Holm $p = 0.001$, $= 0.001$). For $t \leq 4$ its excess risk is within a factor 1.23 of Bayes. After that it decays more slowly than $1/t$ and settles about 1.3×10^{-3} above Bayes. Meanwhile $t\,\mathcal{R}_t^\star$ reaches 0.057 at $t{=}47$: Theorem 1 gives 0.058 and Corollary 2 predicts $np\sigma^2{=}0.06$. The pre-registered tolerance H1c is **rejected**: the mean excess ratio is 1.59 [1.56, 1.62], against a threshold of 1.25. This is the dynamical analogue of the observation that trained transformers fall short of least-squares rates (Hill et al., 2025; Cole et al., 2025); §8 tests whether longer training or larger models close the gap.

The pre-registered mechanism is rejected: no linearly readable, causally used estimate in the residual stream. The best linear probe of the residual stream for $\hat{\theta}_t$ reaches only $R^2{=}0.19$ [0.17, 0.21] (H2a, threshold 0.8: rejected). It beats the probe for the true θ by just 0.005 (H2b: significant but negligible). The normalised sufficient statistics are partly decodable ($R^2{\leq}0.30$) and peak at layers 1–2, before $\hat{\theta}$ peaks at layer 4, in every seed. So H2c holds formally but is uninformative. What *is* linearly decodable is the model's uncertainty: $\log \operatorname{tr} P_t$ reaches $R^2{=}0.66$. Steering along the $\hat{\theta}$-probe directions does not move the prediction (slope -0.0000; H3a rejected). Swapping the 6-d probe subspace of the query between episodes transfers $\tau{=}0.0001$ of the behaviour, which is less than a random 6-d subspace (0.067; H3b rejected). Whole-stream patching (Fig. 1c) shows that parameter information enters the query position only in blocks 3–4: $\tau_{\text{last}}{=}0.38 - -0.62$ after block 3, while patching all context positions at layer 2 still transfers ≥ 0.91.

Exploratory: the sufficient statistics live in the key–value memory and act through the query Jacobian. These analyses were added after the pre-registered ones failed; Remark 4 says where to look, and Appendix E gives the dimension-matched controls. *The answer is linearly represented before the parameter is:* a linear probe recovers the Bayes *prediction* $\hat{\theta}_t z_t$ with $R^2{=}0.88 - -0.92$ already after block 3 (after block 4 this is trivial, the unembedding being linear), while an MLP probe recovers $\hat{\theta}_t$ itself to 0.72: the parameter is present, nonlinearly. *The functional estimate is the Jacobian:* the predictor is near-linear in its query (counterfactual queries are explained with $R^2{=}0.87 - -0.96$ by its Jacobian, slope 0.97–0.99 against the Bayes map), and $J_t{=}\partial\hat{y}_t/\partial z_t$ matches

$\hat{\theta}_t^{\text{Bayes}}$ with R^2=0.92, more closely than the truth (0.89) or OLS (0.90) (Fig. 2a). *The sufficient statistics are in the key–value memory of blocks 3–4:* the per-head fast weights $\widehat{\text{Cov}}(v, k)$ of block 4 predict $\hat{\theta}_{47}$ with R^2=0.88 – −0.89 (block 3: 0.78–0.79), the residual stream at the query position at most 0.08–0.14 (Fig. 2b). This is not a matter of dimension (Table 4): projected to 128 dimensions the fast weights still give R^2=0.84 – −0.86 (PCA: 0.87), while the residual stream gives at most 0.58–0.61 whichever way it is read (query position, all 48 positions concatenated, mean-pooled, or last context token; Table 4). The memory holds the normalised sufficient statistics (S_t, C_t) as directly as the estimate: by block 4 both are linearly readable from the fast weights (R^2=0.92 and 0.90–0.91; residual stream over all positions: 0.50–0.53 and 0.33–0.37), and in the earlier blocks the statistics lead (block 2: 0.70–0.79 against 0.43–0.49). The pre-registered "statistics first, solve later" picture was therefore right about the order and wrong about the location: the statistics accumulate in the key–value memory, not in the residual stream. *The memory is causal and localised:* replacing only the context keys and values of blocks 3–4 by another episode's, with the query's residual stream, its attention queries and all MLPs untouched, transfers τ=0.91 – −0.94 of that episode's behaviour; the same operation on block 1 alone transfers 0.07–0.16, keys alone 0.44–0.48, values alone 0.39–0.63 (Fig. 2c). That context information transfers through keys and values is expected; what the experiment adds is *where*: blocks 3–4, and within them 2–3 near-uniform heads per block, as in Remark 4. Shuffled labels give R^2= − 0.01 – −0.01; the mean-pooled and last-context-token residuals give 0.58–0.61 and 0.18–0.25 for $\hat{\theta}$; and the true statistics predict the estimate *linearly* only with R^2=0.46 (an MLP: 0.87). The memory does better than that (0.90–0.91): the key–value features of blocks 3–4 are shaped so that their covariance across the context is itself a linear image of the estimate, which is exactly what Remark 4 requires of a head that applies $\hat{\theta}$ to the query. The first-order picture of Remark 4 explains R^2=0.84 – −0.96 of the outputs of the near-uniform heads (entropy ≥ 0.8) in blocks 3–4, and fails, as it must, for the 1–3 sharp heads there. Patching the keys and values of the two highest-transfer heads per block gives τ=0.78 – −0.82, against 0.41–0.51 for two random heads. *Probe directions are not the model's directions:* even for the well-decoded prediction, steering along the probe has slope 0.08 and subspace patching transfers 0.08. In short, the transformer implements near-Bayes system identification without a linearly readable $\hat{\theta}$ in the residual stream: the sufficient statistics are held in the key–value memory and applied to the query by attention. This is one reason why RMA-style "latent parameter" probes of activations (Kumar et al., 2021; Chamachot, 2026) can fail even when adaptation works.

6. Experiment B: nonlinear and contact physics in MuJoCo

Prediction. Averaged over t=24..47 (curves in Fig. 4, appendix), the normalised error on the pushed block with Coulomb friction is 0.0058 for the transformer and 0.019 for physics-feature RLS (H4a: difference 0.0131 [0.0128, 0.0134], Holm $p = 0.001$). For comparison, the no-context MLP gets 0.019, nominal-parameter simulation 0.075, and simulation with the true parameters 0.00026. Feature RLS is linear in its parameters and cannot represent stick–slip, so we add a structured identifier with the true model class: a grid search over (m, μ) with the exact simulator, choosing at each t the parameters that best explain the context so far. It reaches 0.011, no better than the exact simulator restarted from the observed state with the *true* parameters (0.011): from $(x, \dot{x})$ alone, one-step prediction is limited by the hidden contact state, which the transformer evidently infers from its history (0.0058). Feature RLS does adapt (its coefficients correlate with $1/m$ and μ at r=0.58 and 0.46); its plateau coincides with the no-context MLP because that is the linear model class's limit.

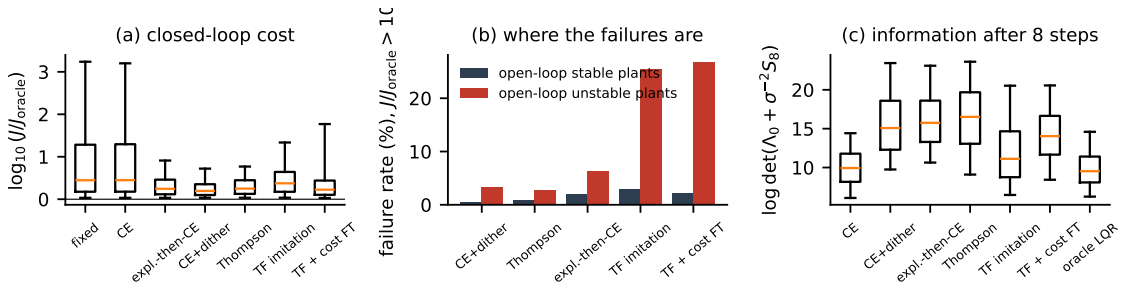


Figure 3: **C** (4096 paired test plants, 3 seeds). (a) Normalised cost J/J_{oracle}; boxes span the 5–95% range. Certainty equivalence without excitation never moves u, so it learns nothing (Prop. 3). (b) Failure rate on open-loop stable vs. unstable plants. (c) Information gathered in the first 8 steps.

On the pendulum, where feature RLS has the right model class up to discretisation, the transformer (3.5e-05) and RLS (5e-05) are on par at the noise scale (true-parameter simulation: 8.9e-06). Nominal simulation (0.002) and the no-context MLP (0.0019) are $\approx 58\times$ worse, so the gain comes from in-context adaptation.

Mechanism. The pre-registered probes again fail for the parameters that enter the dynamics multiplicatively. Residual-stream R^2 at $t{=}47$ is 0.82–0.86 for g, 0.32–0.44 for $1/m$ and 0.51–0.52 for μ (H4c rejected). Steering along the $1/m$ probe moves the input sensitivity the right way in only 42% (pendulum) and 71% (block) of episodes (H4d rejected). The exploratory analyses of §5 transfer. The model's own sensitivity $\partial\dot{q}_{t+1}/\partial u_t$, its functional estimate of $1/m$, correlates with the true $1/m$ at $r{=}0.99 - -0.99$ on the pendulum; on the block it is $r{=}0.42 - -0.43$, because the sensitivity is zero while stuck. The key–value fast weights decode $1/m$ with $R^2{=}0.97 - -0.98$ (residual: 0.32–0.44), and on the block they decode μ at 0.68–0.72 (residual 0.51–0.52).

Out-of-sample check: a bouncing ball. This system was added after the key–value account had been formed: a ball with unknown mass, gravity and contact damping ratio ζ, pushed by random vertical forces. The transformer reaches 0.0003 against 0.0027 for nominal-parameter simulation, 0.0021 for the no-context MLP and 0.037 for a spring–damper feature RLS, which is worse than persistence (0.023): a linear contact model is the wrong class. The pattern of §5 repeats only in part. The key–value memory reads the damping ratio better than the residual stream at the same depth (ζ: 0.98 vs. 0.88–0.89), but $1/m$ is weakly readable from both (0.48 vs. 0.42–0.45), and steering along the $1/m$ probe is inert (39% correct sign). We read this as support for the inertness of residual probes, and as weak support only for the key–value localisation on this system.

7. Experiment C: does an in-context controller explore?

The dual effect in the baselines. With the prior mean $B{=}0$, certainty equivalence (CE) computes a zero gain, outputs $u \equiv 0$, and so never learns B. Its information after 8 steps stays at the prior level (10.0 nats, against 9.9 for the oracle, which needs none), it fails on 30.2% of plants, and its median cost is 2.81. A constant gain cannot help when the actuation direction is unknown (median

2.80). Any excitation fixes this: decaying dither (median 1.57, 1.4% failures, 15.6 nats), explore-then-commit (1.76, 3.4%, 16.1) and Thompson sampling (1.78, 1.4%, 16.4).

The transformer explores, without being told to. Pre-registered pure BPTT did not train (Appendix, Table 1); the controller is a DAgger warm start followed by cost fine-tuning. After the warm start it gathers 12.0 nats; cost fine-tuning adds 2.19 [2.06, 2.37] more, to 14.2. That is 4.20 [4.06, 4.31] above CE (H5d, Holm $p = 0.001$). So a controller with no identification objective learns to excite the plant, and most of that excitation is learned from the cost, not copied from the LQR teacher. It still excites less than dither (-1.41 [-1.55, -1.29]) and Thompson sampling (-2.18 [-2.32, -2.05]).

Cost: competitive in the typical case, unsafe in the tail. The median normalised cost is 1.69 (seeds 1.53–1.86), against 1.57 for dither and 1.78 for Thompson sampling; per plant, the transformer wins against Thompson sampling 51% of the time and against dither 43% (paired median differences 0.01 [-0.09, 0.11] and -0.10 [-0.22, -0.00]). But its tail is heavier: CVaR_{95}, the mean cost of the worst 5% of plants, is 7612 against 11.7 for dither and 11.4 for Thompson sampling, and it fails on 10.0% of plants against 1.4%. The failures sit almost entirely on the 31% of plants that are open-loop unstable: 27% there against 2.3% on stable plants (dither: 3.3% and 0.5%), Fig. 3b. Controllability is not the issue. Because the mean is dominated by these failures, the transformer beats the constant gain (H5a, Holm $p = 0.015$) but *not* CE on the pre-registered mean (H5b: 111.60 [-149.21, 343.32], $p = 1.000$), and it is far worse than every excited baseline in the mean (vs. dither: -379.87 [-648.12, -152.71]).

What cost fine-tuning buys, and what it costs. Relative to the warm start, fine-tuning on $\log(J/J_{\text{oracle}})$ improves the median (2.38 to 1.69) and the excitation, leaves the failure rate unchanged (9.9% to 10.0%) and makes the failures more severe (mean 22.69 to 382.19, difference -359.50 [-626.51, -130.78]): the log objective compresses the tail. Objectives that weight the tail do not fix this: fine-tuning the same warm start on the raw cost or on the CVaR of the worst 10% of plants diverged within 500–1000 steps (validation median above 18 and failures above 67% for raw cost; median above 19 and failures above 74% for CVaR by step 1500), and validation-based selection returned the warm start itself in both cases (Table 1). The catastrophic plants dominate any objective that does not compress them.

Exploratory: what the controller identifies, and where. On its own closed-loop trajectories, the controller's feedback Jacobian $-\partial u_t/\partial x_t$, its functional gain, approaches the oracle LQR gain as the context grows (R^2=0.42 − −0.57 at t=47), but it extracts less from its data than a classical estimator would: the certainty-equivalent gain computed from the *same* history is 0.07 from the oracle gain in squared error, against 0.59 for the transformer. The actuation matrix B is readable from the key–value fast weights with R^2=0.69 − −0.81 already at t=8 (0.65–0.75 at t=47), against 0.47–0.54 (0.39–0.43) from the residual stream at the query position, with the dimension caveat of Appendix E. As in §5, the residual-probe directions are inert: flipping the probed estimate flips the action in 0% of cases.

8. Scale, training length and seeds

Table 3 (appendix; one seed per row, exploratory) asks whether the approximation floor of §5 is an artefact of a small model or short training, and whether the Jacobian account survives a larger

system. *The floor is partly an optimisation and capacity limit.* Tripling the training steps lowers the mean excess ratio to Bayes from 1.59 to 1.22 (at $t{=}47$ from 2.04 to 1.38), doubling the depth gives 1.23 and quadrupling the width 1.30, while a 0.1M-parameter model is far off (5.27); none reaches the Bayes risk. *The functional account holds across sizes:* at $t{=}47$ the query Jacobian matches the ridge estimate with $R^2 \geq 0.99$ for the long-trained, deep and wide models (small model: 0.89). *On a larger system* ($n{=}4$, $m{=}2$, 24 unknowns) the transformer is far from Bayes (excess ratio 4.4) yet its Jacobian still matches the ridge estimate ($R^2{=}0.96$ at $t{=}47$); the Bayes risk follows the finite-sample form of Corollary 2, $t\,\mathcal{R}^\star_{47}{=}0.276$ against $np\sigma^2\,t/(t{-}p{-}1){=}0.282$. *Seeds.* Seven further training seeds of the main model, trained after the key–value account was formed, replicate the behaviour (mean excess ratio 1.58 ± 0.03, mean $\pm$ s.d. over 10 seeds), the localisation (KV-only patching of blocks 3–4 transfers 0.93 ± 0.01; minimum over seeds 0.91), and the readability gap: the fast weights read $\hat{\theta}_{47}$ with 0.91 ± 0.01 (minimum 0.90), the residual stream with 0.24 ± 0.11 at the query and 0.37 ± 0.07 over all positions; the residual readout varies more across seeds and reaches 0.55 in the best one.

9. Discussion and limitations

What the transformer computes. In all three settings the trained models are competent in-context system identifiers, but their *form* of identification is not the one we pre-registered: there is no linearly readable, causally used parameter vector in the residual stream (an MLP probe recovers part of it, so the information is present nonlinearly). The sufficient statistics of the context sit in the key–value memory of the last two blocks, where they are read with $R^2{=}0.92$ against at most 0.68–0.71 from the residual stream (query, all positions, mean-pooled or last context token), and the model's query Jacobian, rather than any activation, is what matches the Bayes estimate. For interpretability of adaptive policies, probe the *function* (Jacobians) and the *memory* (keys and values), not only the activations.

Pre-registration. The pre-registration is a commit in the repository's history (tag `prereg-d134c5a`), not an externally time-stamped record. Its thresholds and causal tests (H3a/H3b) are what rejected the tempting reading of a probe with $R^2{=}0.19$ as "the model represents its estimate".

Limitations. (i) *Scale.* The models have 0.2–3M parameters, the linear systems have at most 24 unknowns, and the MuJoCo systems have one degree of freedom. Whether the key–value account holds for large robot policies is open. (ii) *Post-hoc mechanism.* The key–value account was formulated after the pre-registered probes failed, on the same models. The bouncing-ball system, trained after the account was formed, reproduces the inert residual probes but supports the key–value localisation only for one of three parameters (§6). (iii) *Seeds.* Three training seeds per model; the cluster bootstrap over three seeds under-estimates between-seed variance, so the confirmatory p-values are conditional on these seeds, and seed-level results are reported alongside. (iv) *Approximation floor.* The predictors stay a factor 1.59 above the Bayes risk averaged over contexts (H1c rejected). Tripling the training steps or doubling the depth brings it to about 1.22, not to 1 (one seed each). (v) *Control recipe.* Pure closed-loop BPTT, as pre-registered, did not train on open-loop-unstable plants; the controller results are for a DAgger warm start followed by cost fine-tuning on a log objective; fine-tuning on the raw cost or on its CVaR diverged. (vi) *Baselines in MuJoCo.* Feature RLS is linear in its parameters and cannot represent stick–slip; the structured identifier of §6 is the model-based reference there. (vii) *Authorship.* Single-author work with substantial AI assistance (see below).

Generative AI use

Generative AI tools (Claude) were used for literature search and synthesis, code implementation, mathematical derivations, and writing. The author reviewed the content, every reported number is regenerated by the scripts in the public repository, and the author takes responsibility for the final content.

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Appendix A. A construction that gathers the sufficient statistics

Proposition 5 (Two attention layers suffice to gather the sufficient statistics) *Let token t be $e_t = (z_t, 1, \pi_t)$, with $\pi_t \in \{0,1\}^T$ a one-hot position. There is a two-layer causal* linear-*attention network (one head in layer 1, p heads in layer 2, no MLPs) whose residual stream at every position t contains S_t and C_t* exactly*, together with z_t and t. Consequently, the Bayes predictor of Theorem 1 is a fixed continuous function $\phi(S_t, C_t, z_t)$ of the token-t residual alone. A token-wise MLP approximates it uniformly on any compact set, so in this construction all further depth is free for the "solve" ϕ rather than for gathering data.*

Proof Appendix B: a previous-token head, then p constant-query linear-attention heads. ■

Proposition 5 is in the spirit of earlier constructions (Akyürek et al., 2023; Bai et al., 2023; von Oswald et al., 2023). Its role is to make a *falsifiable* mechanistic prediction for trained models. If a trained transformer works this way, a linear probe should decode the normalised statistics

$(S_t, C_t)/(t{+}1)$ no later in depth than it decodes the posterior mean $\hat{\theta}_t$ (H2c). If it instead refines $\hat{\theta}$ iteratively in the style of GD (von Oswald et al., 2023), $\hat{\theta}$ should become decodable gradually, with no early peak for the statistics.

Appendix B. Omitted proofs

Proof of Theorem 1. The likelihood of $\mathcal{D}_t$ factorises into the policy factors $p(u_s \mid \mathcal{D}_s)$, which do not depend on θ, and $\prod_{s<t} \mathcal{N}(x_{s+1}; \theta z_s, \sigma^2 I)$. Hence the posterior of each row is Gaussian, $\theta_i \mid \mathcal{D}_t \sim \mathcal{N}(m_i, \Sigma_t)$, the rows are independent, and $\Sigma_t = (\Lambda_0 + \sigma^{-2} S_t)^{-1} = \sigma^2(\sigma^2 \Lambda_0 + S_t)^{-1}$ and $[m_1, \ldots, m_n] = \Sigma_t \, \sigma^{-2} C_t$. Since u_t is conditionally independent of θ given $\mathcal{D}_t$, the posterior given $(\mathcal{D}_t, u_t)$ is the same. For any $(\mathcal{D}_t, u_t)$-measurable $\hat{y}_t$, the bias–variance identity gives $\mathbb{E}[\|\hat{y}_t - \theta z_t\|^2 \mid \mathcal{D}_t, u_t] = \|\hat{y}_t - m z_t\|^2 + \sum_i z_t^\top \Sigma_t z_t$. Taking expectations yields (1), with equality iff $\hat{y}_t = m z_t$.

Proof of Corollary 2. Let $M_t = t(\sigma^2 \Lambda_0 + S_t)^{-1}$, so $M_t \to \Sigma_\theta^{-1}$ a.s. by (a). Then $|z_t^\top (M_t - \Sigma_\theta^{-1}) z_t| \le \|z_t\|^2 \|M_t - \Sigma_\theta^{-1}\|$, which tends to 0 in probability because $\|z_t\|^2$ is tight by (c). The difference is uniformly integrable as a difference of uniformly integrable families, so its expectation vanishes. Hence $\mathbb{E}[z_t^\top M_t z_t \mid \theta] = \mathrm{tr}(\Sigma_\theta^{-1} \mathbb{E}[z_t z_t^\top \mid \theta]) + o(1) \to p$ by (b). Multiply by $n\sigma^2$. The unconditional statement follows by dominated convergence.

Proof of Proposition 5. *Layer 1 (previous-token head).* Set $q_t = \Pi \pi_t = \pi_{t-1}$, with Π the down-shift and $\pi_{-1} = 0$; set $k_s = \pi_s$ and $v_s = z_s$. Linear attention outputs $\sum_{s \le t} v_s \, k_s^\top q_t = v_{t-1}$. The residual becomes $(z_t, 1, \pi_t, z_{t-1})$, with $z_{-1} = 0$. *Layer 2 (moment heads).* Head $j \in \{1..p\}$ uses the constant query $q_t = 1$, the key $k_s = (z_{s-1})_j$, and the value $v_s = (z_{s-1}, x_s)$. It outputs $\sum_{s \le t} (z_{s-1}, x_s)(z_{s-1})_j$, which is the j-th column of $[S_t; C_t^\top]$, because the pairs (z_{s-1}, x_s), $1 \le s \le t$, are exactly the context pairs. Writing the p head outputs to disjoint residual coordinates gives S_t and C_t. The map ϕ is continuous on $\{\sigma^2 \Lambda_0 + S \succ 0\}$, so the universal approximation theorem applies on compacta.

Appendix C. Deviations from the pre-registration

(1) We measure excess risk against the noiseless mean θz_t instead of as MSE $- n\sigma^2$. The two have the same expectation and ours has lower variance. (2) $\Lambda_0{=}I$ in the information measure, which the pre-registration did not fix. (3) Training details not fixed by the pre-registration: the constant-gain baseline uses Adam, 1000 steps and batch 1024; the no-context MLP has $2{\times}256$ units and 5000 steps. (4) The controller uses a KV-cached incremental forward to fit in 4 GB. It is numerically identical to the full forward. (5) With a prior mean $B{=}0$, pre-registered CE degenerates to $u \equiv 0$; we report it as specified. (6) On the pendulum the parameter effects are small (nominal-parameter simulation already has 0.002 normalised error); we kept the design unchanged. (7) All analyses labelled *exploratory* were added after the pre-registered probes failed: the query Jacobian, fast-weight probes, KV-only patching, answer probes, MLP probes, the bouncing-ball system, the Thompson-sampling baseline and the scale study. (8) *Controller training.* Pure closed-loop BPTT on the cost, as pre-registered, was unstable in pilots (Table 1). The final controller uses a DAgger warm start (imitating the oracle LQR action from its own history) followed by BPTT fine-tuning on $\log(J/J_{\text{oracle}})$, with 8-step truncation and learning rate 10^{-4}, for the pre-registered 6000 BPTT steps. (9) Execution only: BPTT is replayed as a CUDA graph; the constant-gain baselines are trained on the CPU. The full log is in `docs/DEVIATIONS.md`.

Appendix D. Implementation details

Exact Bayes predictor (A). Each row of θ has the Gaussian posterior of Theorem 1, and the rows share their covariance. We draw 2048 joint samples, discard those with $\rho(A) \geq 0.95$, and average. The acceptance rate is 0.94 at t=0 and 0.996 at t=47. **Probes.** Ridge regression on standardised residuals, with α chosen by 5-fold CV grouped by episode, fitted on 2048 test episodes and scored on the other 2048. **Steering.** Let D be the probe readout matrix. The minimal-norm residual change that moves readout coordinate k by δ while leaving the others unchanged is $\delta\,(D^{+})_k$. **Patching transfer** is $\tau = \langle \hat{y}_{\text{patch}} - \hat{y}_B, \hat{y}_{A'} - \hat{y}_B \rangle / \|\hat{y}_{A'} - \hat{y}_B\|^2$, where A' is episode A's context followed by episode B's query token. **Compute.** Everything runs on a laptop RTX 3050 Ti (4 GB) with a 20-thread CPU, in about 15 GPU-hours end to end.

Table 1: Controller-training pilots (seed 99, 1024 held-out plants). Cost is J/J_{oracle}; "fail" means J/J_{oracle}>10. †Intended lr 10^{-4}, but a bug ran these pilots at $3{\cdot}10^{-4}$. The final runs use 10^{-4}.

variant	step	median	mean	fail
BPTT, J/J_{oracle}, lr $3{\cdot}10^{-4}$	1000	10.49	298	51%
	1999	84.89	1146	96%
BPTT, $\log(J/J_{\text{oracle}})$, lr $3{\cdot}10^{-4}$	1000	2.41	300	21%
	1999	11.67	976	52%
BPTT, log, squashed inputs†	1000	137.67	76387	98%
	1999	156.09	51593	98%
same, 8-step truncated BPTT	1000	4.35	6542	36%
DAgger warm start (3000) + fine-tune (1500)†	0	2.69	204	19%
	1000	2.09	1070	24%
	1499	1.99	250	16%
final warm start (6000) + raw cost, seed 0	0	2.40	8	8%
	500	18.07	1366	67%
	1000	59.23	13750	89%
	2999	59.53	1716	90%
final warm start (6000) + $\text{CVaR}_{0.9}$, seed 0	0	2.39	7	9%
	500	2.74	319	14%
	1000	24.15	1801	78%
	2999	19.77	309	74%

Appendix E. Dimension-matched probe controls

Appendix F. Experiment B: curves and parameter readability

Table 2: Seed-level results. A: excess-risk ratio to Bayes (mean over t=4..47) and residual-probe R^2 for $\hat{\theta}$ at the best layer. C: median and mean normalised cost, failure rate and information after 8 steps, for the imitation-only warm start and the cost-fine-tuned controller.

	A		C, imitation only				C, imitation + cost			
seed	ratio	$R^2(\hat{\theta})$	median	mean	fail	info	median	mean	fail	info
0	1.59	0.21	2.35	19	8.3%	12.0	1.53	569	9.2%	14.1
1	1.61	0.18	2.23	26	9.6%	12.2	1.86	420	11.2%	14.3
2	1.57	0.18	2.58	24	11.8%	12.0	1.69	157	9.5%	14.3

Table 3: Scale study (exploratory, one seed unless stated). Ratio: excess risk relative to the Bayes (ridge) predictor, mean over t=4..47 and at t=47. $R^2(J)$: agreement of the query Jacobian with the ridge estimate, mean over $t \in \{4, 8, 16, 32, 47\}$.

model	change	params	ratio ($t \geq 4$)	ratio (t=47)	$R^2(J)$
A (3 seeds)	reference, 20k steps	0.8M	1.59	2.04	0.92
long60k	same model, 60k steps	0.8M	1.22	1.38	0.77
small64	d=64, 2 layers	0.1M	5.27	9.37	0.73
deep8	8 layers	1.6M	1.23	1.34	0.76
wide256	d=256	3.2M	1.30	1.53	0.76
n4m2_s0	n=4, m=2, seed 0	0.8M	4.42	4.32	0.72

Table 4: Dimension-matched probe controls at t=47 (held-out R^2, best block, min–max over 3 seeds). Ridge probes in dual form with 5-fold CV. (S, C) are the normalised sufficient statistics.

features	dim	$\hat{\theta}_{47}$	$(S, C)_{47}$	θ
residual stream, query position	128	0.18–0.25	0.28–0.29	0.18–0.25
residual stream, all 48 positions	6144	0.33–0.37	0.50–0.53	0.32–0.36
random projection of the above to 128	128	0.01	0.08–0.09	0.01
key–value fast weights $\widehat{\mathrm{Cov}}(v, k)$	4224	0.90–0.91	0.92	0.89
random projection of fast weights to 128	128	0.84–0.86	0.75–0.76	0.83–0.84
top-128 PCs of fast weights	128	0.87	0.80–0.81	0.86
residual stream, mean over 48 positions	128	0.58–0.61	0.68–0.71	–
residual stream, last context token	128	0.18–0.25	0.29–0.31	–
fast weights, shuffled labels (control)	4224	-0.01–0.01	–	–
true statistics $(S, C)_{47} \to \hat{\theta}_{47}$, linear / MLP (ceiling)	12	0.46 / 0.87	–	–

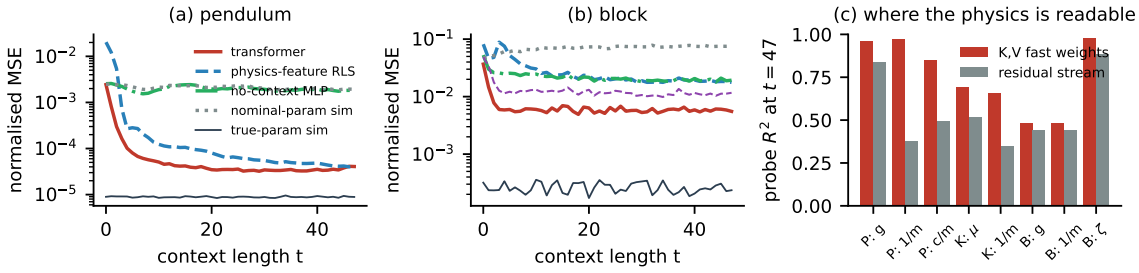


Figure 4: **B** (3 seeds). (a, b) Normalised one-step error vs. context length. On the pendulum the transformer is on par with physics-feature RLS, which there has the true model class up to discretisation. On the stick–slip block it beats RLS by $3.2\times$. (c) Probe R^2 at t=47 for each physical parameter, from the key–value fast weights (best block) vs. from the residual stream (P: pendulum, K: block, B: bouncing ball).